SMART INDIA HACKATHON 2025

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TITLE PAGE

- **Problem Statement ID –** 26207
- **Problem Statement Title-** Student Innovation
- **Theme-** Smart Education
- **PS Category- Software/Hardware** Software
- **Team ID-**
- **Team Name (Registered on portal)** Code Blooded

Code Blooded

Disha — it points, you click, and then it fades

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

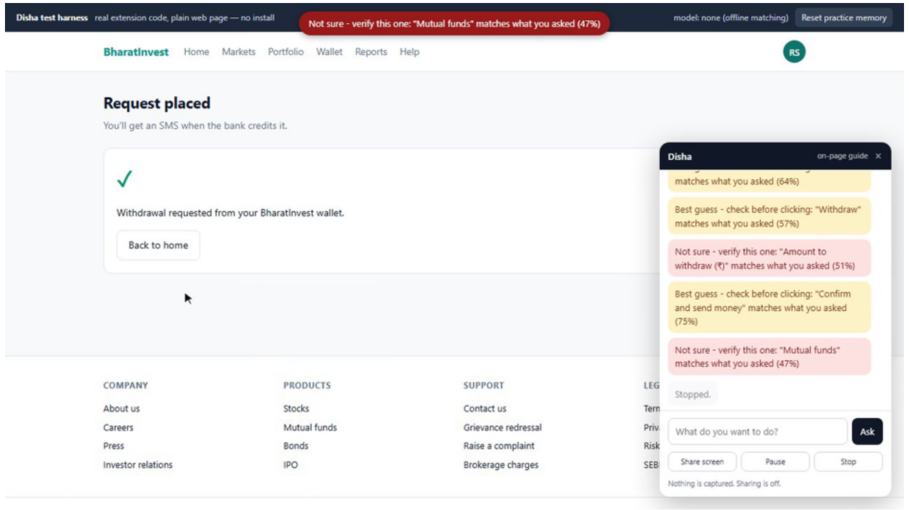
- Disha reads the page's own controls and their accessible names, then marks the one next step with a colour-coded dot.
- **You click. Disha never can.** At an OTP or payment step it pauses and hands control back.
- **Every success dims the dot; five clean repetitions remove it entirely.**
- **Not pre-scripted for any portal** — it reads whatever is on screen.

How it addresses the problem

- **A citizen who does not know the word** withdraw still finishes the task — and finishes it themselves.
- No paid intermediary, no travel, no “come back next week”.
- **Because it points instead of clicking,** it is safe on Aadhaar, banking and scholarship portals where an autopilot agent is not.

Innovation and uniqueness of the solution

- **Confidence you can see.** The dot's colour is not the model's self-report — the page evidence is scored separately and both must agree.
- **It teaches, then leaves.** Guidance fades on success, returns after a mistake.
- **Zero integration cost.** No ministry rebuilds anything.



Live capture — four dots, three screens, one goal typed in plain words.

What the colour means

- **Green:** model ≥ 0.80 and page ≥ 0.75
- **Amber:** best guess — check before clicking
- **Red:** not sure — verify this one

Closest shipped product: Microsoft Copilot Vision “Highlights.” We do not claim to have invented on-screen pointing. Ours is the combination — exposed confidence, DOM + ARIA grounding, a fading scaffold, and India-specific task packs.

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TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Manifest V3 Chrome extension**, vanilla JavaScript, no build step.
- **Model is pluggable** — Gemini free tier, Anthropic, or a local Ollama model with no internet.
- **DOM + ARIA**, not Chrome's accessibility tree — that needs the debugger permission and breaks on old government markup.
- **Vision grounding (GUI-Actor, MIT)** is the documented fallback.

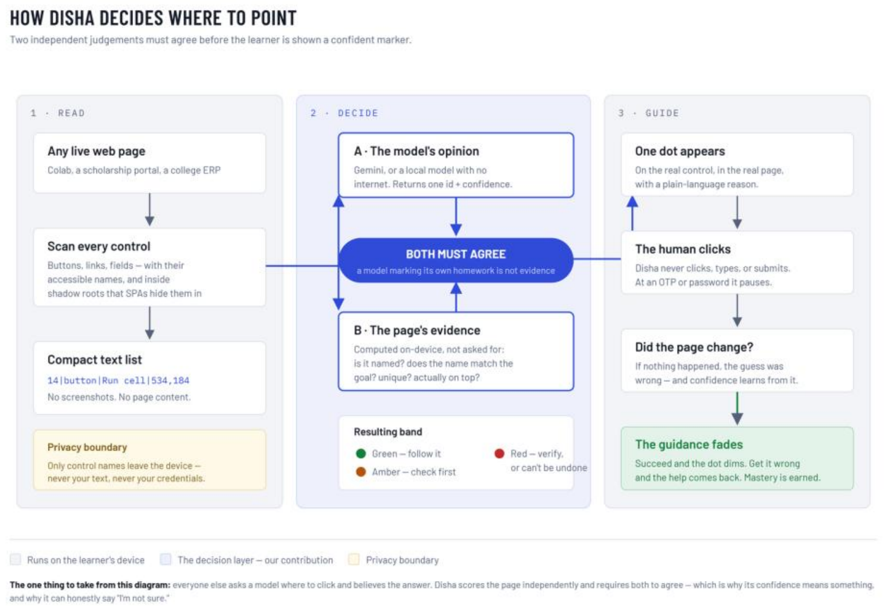
Methodology and process for implementation (Flow Charts/Images/ working prototype)

```
page → enumerate interactive elements + ARIA names → compact text list → model returns {id, confidence} → on-device grounding score → colour band → draw dot → human clicks → wait for the DOM to settle → next step
```

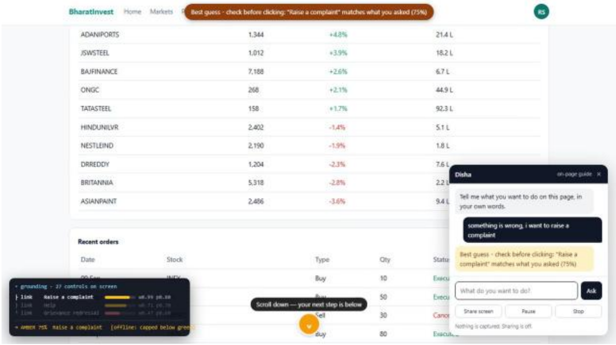
- **Green needs 0.80 model** and 0.75 page evidence — two numbers, not one.
- **We display the weaker of the two, never the average** — so the number can never contradict the colour.

Product status

- **Working prototype, verified end-to-end in a browser:** live grounding, colour bands, contingent fading, on-page ask bar, screen share with a real pause, offline mode. Vision fallback and the live-model green dot are next.



Read → Decide → Guide, and the boundary nothing crosses: only control names leave the device.



Every navigation state in one take — two arrowheads, then one, then the dot, then the same going up.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

- **Technical:** built and running today; DOM + ARIA works on portals we do not control.
- **Financial:** a compact text request per step on a free-tier model; a local model costs zero. Cloud for the next build is already covered — $1,000 in Microsoft Azure credits through Red Bull Basement 2026.
- **Operational:** installs as an extension. No ministry integration, no portal changes.

Potential challenges and risks

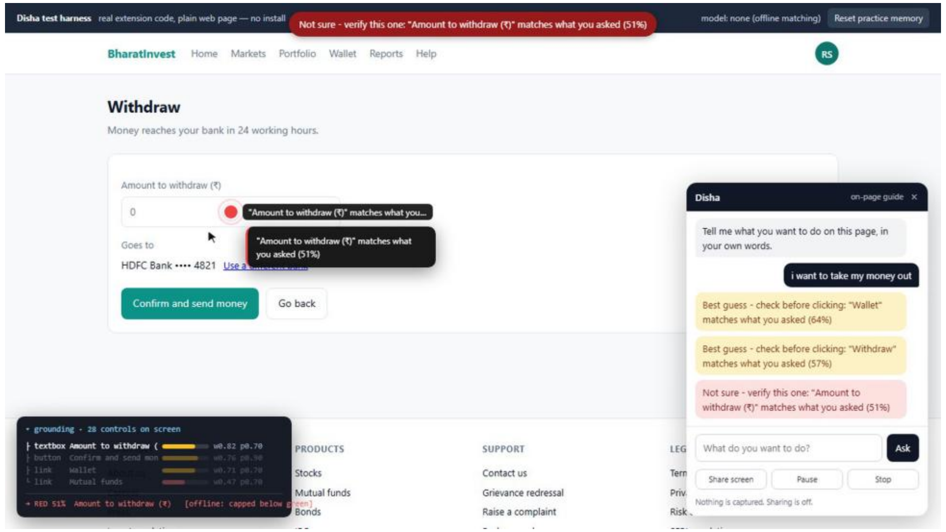
- **Unlabelled / icon-only controls** — our weakest case.
- **The model picks the wrong element.**
- **Trust:** users may over-rely on a confident-looking dot.
- **Market:** Copilot Vision already points at things.

Strategies for overcoming these challenges

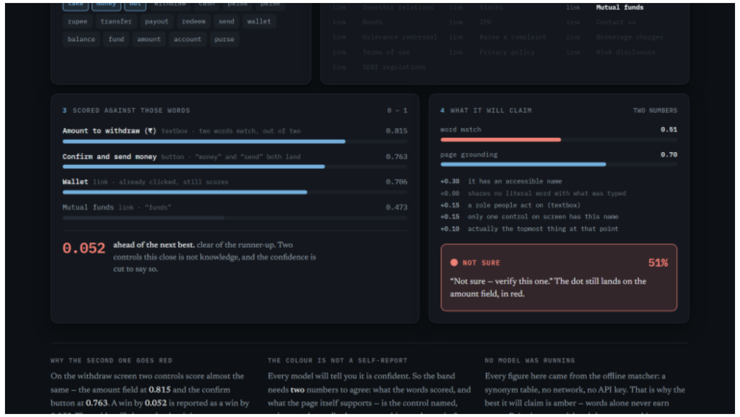
- **Unlabelled controls drop the page score → the dot goes red**, never a silent guess; vision grounding is the fallback.
- **A wrong pick costs a glance, not a transaction** — the human always clicks.
- **Over-reliance is answered by the fade itself.**

The measured run

- **“i want to take my money out” on our practice portal:** 64% → 57% → 51% red → 75%.
- **That run used the offline matcher with no model,** which is capped below green by design. Accuracy is unmeasured.



The near-tie that goes red at 51%, and the card that says why.



Both numbers, side by side: page 0.815, words 0.763, margin 0.052.

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IMPACT AND BENEFITS

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Potential impact on the target audience

- **Positive — improvement:** the citizen completes the task alone instead of paying someone to click.
- **New opportunities:** a teacher or NGO worker walks a task once and it becomes a pack others reuse.
- **Negative — technology adoption:** it needs a browser and a first install.

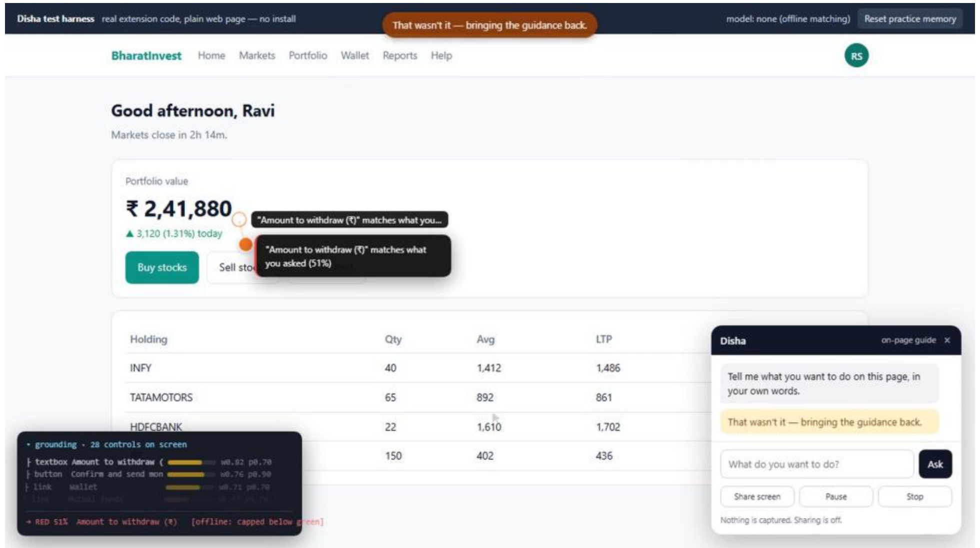
Benefits of the solution (social, economic, environmental, etc.)

- **Social — improved access, empowerment:** independence on services people are already entitled to.
- **Economic — cost:** avoids the per-visit intermediary and travel documented in the CSC studies.
- **Educational:** guidance fades on demonstrated success — active learning +0.47 SD, spaced retrieval g = 0.74. Both validate the mechanism, not our product.

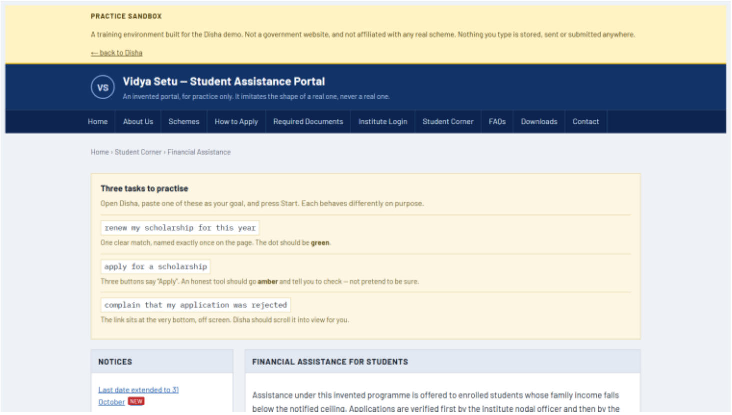
21%
of rural Indians aged 15–24 can search, email and bank online
MoSPI CAMS 2022–23, combined ICT-skill measure

11.4%
of connected households use the internet for government services
NCAER, June 2026

1 in 5
households using digital services depends on help from outside
NCAER, June 2026



The same task three times until the dot is gone — the education claim, on video.



Vidya Setu — the practice sandbox, shaped like a government portal, safe to fail in.

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

- **MoSPI**, Comprehensive Annual Modular Survey 2022–23 — the 21% combined ICT-skill figure.
- **NSS 78th Round (2020–21)** — computer literacy 24.7% national, 18.1% rural.
- **NCAER**, The Evolving Landscape of Digital Inclusion in India, June 2026.
- **PIB / MeitY** — CSC network: 48.54 crore transactions, 5,01,731 centres, FY 2025–26.
- **IAMAI–Kantar**, Internet in India 2024 · BHASHINI metrics dashboard.

- **Freeman et al., PNAS 2014** — active learning, +0.47 SD across 225 studies.
- **Adesope et al. 2017** (retrieval, g = 0.51) · **Latimier et al. 2021** (spacing, g = 0.74).
- **Renkl et al.** — guidance fading triggered by mastery evidence, not elapsed time.
- **Microsoft Copilot Vision release notes** — the closest shipped prior art.
- **microsoft/GUI-Actor** (MIT) — the vision-grounding fallback on our roadmap.

What these sources do and do not say

- **The two learning figures validate the mechanism, not our product.** We have not measured Disha's accuracy, and we do not quote one.
- **48.54 crore is a count of transactions, not of citizens;** 21% is a combined ICT-skill measure, not “cannot use a government website”.

Our own work, and where else this idea has been

- **github.com/Saurabh0003M/SIH** — the working prototype, the practice portal, the recorded runs, and ATTRIBUTION.md: every borrowed repository with its licence and commit hash.
- **Read-only grounding runs on live public portals** (scholarships.gov.in, swayam.gov.in, aicte-india.org) are recorded in deck/real-site-run.md — including the ones Disha got wrong and reported red.
- **Red Bull Basement 2026:** this idea is in the programme. Participating teams receive $1,000 in Microsoft Azure credits through Microsoft for Startups — an application-phase benefit, not a prize.